3. $U \times U^T$ U

3a) h^2

On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh disciplinary issues.

1) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$

1) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 14 & -6 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \mp 1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}^2$

$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 0 & 14-\alpha & -6+\gamma & 2-3\alpha \end{bmatrix}$

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Satır ya da sütunları istenen sayıyla çarpıp, toplayarak alttaki matrislere eşit hale getirip eşitleyip denklem çözerek bilinmeyenleri bulma

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-2\alpha \\ 0 & 14-\alpha & -6+\gamma & 2-3\alpha \end{bmatrix}$$

$$\begin{bmatrix} 1 & 14 & -6 & 12 & 2-2\alpha \\ 3\alpha & 5 & -9\gamma & 1 & \\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-2\alpha & \\ 0 & 14-\alpha & -6+\gamma & 2-3\alpha & \end{bmatrix}$$

$$\begin{bmatrix} 1 & 14 & -6 & -12-2\alpha \\ 3\alpha & 5 & -9\gamma & -4 \\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-\beta \\ 0 & 14-\alpha & -6+\gamma & -12-2\alpha \end{bmatrix}$$

$$\begin{bmatrix} 1 & x & -\gamma & 0 \\ 3\alpha & 5 & -9\gamma & -4 \\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-\beta \\ 0 & 14-\alpha & -6+\gamma & -12-2\gamma \end{bmatrix}$$

$$\begin{bmatrix} 1 & x & -\gamma & 0 \\ 3x & 5 & -9\gamma & -4 \\ 0 & \beta-2\gamma & 15 & \alpha^2+24+4x+\beta \\ 0 & 14-\alpha & -6+\gamma & -12-2\alpha \end{bmatrix}$$

$$\begin{bmatrix} 1 & \alpha & -\gamma & 0 \\ 0 & 5-3\alpha^2 & -9\gamma+3\alpha\gamma & -4 \\ 0 & \beta-2\gamma & 15 & x^2+24+4x-\beta \\ 0 & 14-\alpha & -6+\gamma & -12-2\alpha \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$$

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(3) a) $H^2 = (I - 2uu^T)^2 = I - 4uu^T + 4uu^T uu^T$

$u^T u = 1$ $\quad H_2 = I$ $\quad H$ orthogonal matrix

$H = H^T$

$H^T H = 1$

b) $u = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}$ $\quad |u| = 1$ $\quad \sum_{k=1}^{n} a_k^2 = 1$

$|u^2| = 1$

$\lambda = 1$

$H_{m \times m}$

$H^2 = (I - 2UU^T)(I - 2UU^T)$

$I^2 - 2I(UU^T) - 2(UU^T)I + 4(UU^T)(UU^T)$

$H^2 = I^2 - 2I(UU^T) - 2(UU^T)I + 4U(U^T U)\cdot U^T$

$I - 2UU^T - 2UU^T + 4U \cdot U^T$ $\quad U^T u = 1$

$I - 4UU^T + 4UU^T =$

I

c) $Hu = (I - 2uu^T)u = u - 2u = -u$

$\lambda = -1$

$H = I - 2UU^T$

$HU = (I - 2UU^T)U$

$HU = U - 2(UU^T)U$

$HU = U - 2U(U + U)$

$HU = U - 2U(U^T U) = 1$

$HU = U - 2U$

$HU = -U$

So an eigen vector of H is u itself, corresponding eigen value $= -1$

Let this -1 be an eigen value of H with its corresponding eigen vector U.

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Difüzyon işleminin $(V-w)$ orantılı olduğunu göz önüne alırsak

(4) (a)

$$\frac{dw}{dt} \propto (v-w) \qquad \frac{dv}{dv} \propto -(v-w)$$

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & +1 \\ +1 & -1 \end{bmatrix} \cdot \begin{bmatrix} v \\ w \end{bmatrix} \qquad \begin{matrix} v(0) = 10 \\ w(0) = 40 \end{matrix}$$

(b) $\frac{dv}{dt} = (-v+w) \qquad \frac{dw}{dt} = v - w$

taking laplace of

$$s\,V(s) - V(0) = -V(s) + W(s)$$
$$s\,W(s) - W(0) = V(s) - W(s)$$

$$= s\,V(s) + V(s) - W(s) = 10$$
$$s\,W(s) + W(s) = -V(s) = 40$$

$$\begin{bmatrix} s+1 & -1 \\ -1 & s+1 \end{bmatrix} \begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} 10 \\ 40 \end{bmatrix}$$

$$\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} s+1 & -1 \\ -1 & s+1 \end{bmatrix}^{-1} = \begin{bmatrix} 10 \\ 40 \end{bmatrix}$$

$$\begin{bmatrix} V \\ W \end{bmatrix} = \frac{1}{s^2+1} \begin{bmatrix} s+1 & 1 \\ 1 & s+1 \end{bmatrix} \begin{bmatrix} 10 \\ 40 \end{bmatrix}$$

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$$\begin{bmatrix} V \\ W \end{bmatrix} = \frac{1}{[(s^2+1)]} \begin{bmatrix} 10s + 10 + 40 \\ 10 + 40s + 40 \end{bmatrix}$$

$$\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} \frac{10s+50}{s^2-1} \\ \frac{40s+50}{s^2-1} \end{bmatrix}$$

$$V_{(s)} = \frac{10s+50}{s^2+2s} \qquad W_{(s)} = \frac{40s+50}{s^2+2s}$$

taking inverse laplace

$$V_{(T)} = 25 - 15e^{-2t} \qquad W_{(T)} = 25 + 15e^{-2t} = 22.96$$

$10 + 40 = 50$

$$\% \frac{22.96}{50} \times 100 = 45.93$$

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